**Solutions Tutorial 6**

The $D$-test:

Define $D(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ .

Suppose $(a,b)$ is a stationary point of $f$, i.e., $f_x(a,b) = f_y(a,b) = 0$.

Then the second partial derivative test ($D$–test) asserts the following:

Case 1. $D(a,b) > 0$.

(a) If $f_{xx}(a,b) > 0$ (or $f_{yy}(a,b) > 0$), then $(a,b)$ is a local minimum.

(b) If $f_{xx}(a,b) < 0$ (or $f_{yy}(a,b) < 0$), then $(a,b)$ is a local maximum.

Case 2. If $D(a,b) < 0$, then $(a,b)$ is a saddle point of $f$.

Case 3. If $D(a,b) = 0$ or none of the above, then the test is inconclusive.

Remarks: For Case 1.

1. The condition $f_{xx}(a,b) > 0$ is equivalent to $f_{yy}(a,b) > 0$; or $f_{xx}(a,b) + f_{yy}(a,b) > 0$.

2. If $f_{xx}(a,b) = 0$ or $f_{yy}(a,b) = 0$, then the test is inconclusive.

$D$-test can be used to determine local extrema, and these local extrema may not be the absolute extrema. To determine the absolute extremum, one may use completing the square (tallying the local extrema point).

Find the stationary points of the following functions and determine their nature:

1. (a) $f(x,y) = x^2 - xy + y^2 = (x - \frac{1}{2}y)^2 + \frac{3}{4}y^2$

Solution. $f_x = 2x - y,\ f_y = 2y - x,\ f_{xx} = f_{yy} = 2,\ f_{xy} = -1.\ D(x,y) = 2^2 - (-1)^2 = 3.$

$f_x = 0 = f_y \Rightarrow 2x - y = 0 = 2y - x \Rightarrow (x,y) = (0,0).$

Since $D(0,0) = 3 > 0$ and $f_{xx}(0,0) = 2 > 0$, it follows that $(0,0)$ is a local minimum.

Alternative. $f(x,y) = x^2 - xy + y^2 = (x - \frac{1}{2}y)^2 + \frac{3}{4}y^2 \Rightarrow f_{\min}(0,0) = 0$ (absolute min).

(b) $f(x,y) = xy(6 - x - 2y) = 6xy - x^2y - 2xy^2$

Solution. $f_x = 6y - 2xy - 2y^2,\ f_y = 6x - x^2 - 4xy,\ f_x = f_y = 0$

$\Rightarrow 6y - 2xy - 2y^2 = 6x - x^2 - 4xy = 0$

$\Rightarrow y(3 - x - y) = x(6 - x - 4y) = 0$

$\Rightarrow y = 0$ or $x + y = 3$; and $x = 0$ or $x + 4y = 6$.

Case 1. $y = 0$ and $x = 0$ implies that $(x,y) = (0,0)$.

Case 2. $y = 0$ and $x + 4y = 6$ implies that $(x,y) = (6,0)$.

Case 3. $x = 0$ and $x + y = 3$ implies that $(x,y) = (0,3)$.

Case 4. $x + y = 3$ and $x + 4y = 6$ implies that $(x,y) = (2,1)$.

Case 5. $y = 0$ and $x + y = 3$ implies that $(x,y) = (3,0)$, a contradiction

(either to $x = 0$ or $x + 4y = 6$ or both).

Case 6. $x = 0$ and $x + 4y = 6$ implies that $(x,y) = (0,\frac{3}{2})$, also a contradiction

(either to $y = 0$ or $x + y = 3$ or both).

| **AND** | $y = 0$ | $x + y = 3$ | $(x,y) = (3,0)$ |
|---|---|---|---|
| $x = 0$ | Case 1 | Case 3 | Contradiction |
| $x + 4y = 6$ | Case 2 | Case 4 | Contradiction |
| $(x,y) = (0,\frac{3}{2})$ | Contradiction | Contradiction | Contradiction |

$f_{xy} = 6 - 2x - 4y, f_{xx} = -2y, f_{yy} = -4x \Rightarrow D = 8xy - (6 - 2x - 4y)^2$

| S.P. $(a,b)$ | $f_{xx}(a,b)$ | $D$ | Nature |
|---|---|---|---|
| (0,0) | | –36 $<$ 0 | Saddle point |
| (0,3) | | –36 $<$ 0 | Saddle point |
| (6,0) | | –36 $<$ 0 | Saddle point |
| (2,1) | – 2 $<$ 0 | 12 $>$ 0 | Local max |

Remark: Note that local maximum point (2,1) will not give $f(2,1) = 4$ the absolute maximum. For example, $f(10,-1) = 20 > 4$.

(c) $f(x,y)=(x^2-y^2)e^{-\frac{1}{2}(x^2+y^2)}$.

Solution. Let $u=\frac{1}{2}(x^2+y^2)$. Then

$$\begin{aligned} f_x &= x(2-x^2+y^2)e^{-u} \\ f_y &= -y(2+x^2-y^2)e^{-u} \end{aligned} \qquad \begin{aligned} f_{xx} &= (2-5x^2+x^4-x^2y^2+y^2)e^{-u} \\ f_{yy} &= (5y^2-2+x^2y^2-x^2+y^4)e^{-u} \\ f_{xy} &= xy(x^2-y^2)e^{-u} \end{aligned}$$

When $f_x(a,b)=f_y(a,b)=0 \Rightarrow x(2-x^2+y^2)=0=y(2+x^2-y^2)$, and since $e^{-u}>0$, it follows that $x=0$ or $x^2-y^2=2$; and $y=0$ or $x^2-y^2=-2$.

Case 1. $(x,y)=(0,0)$.

Case 2. $x=0$ and $x^2-y^2=-2 \Rightarrow (x,y)=(0,\pm\sqrt{2})$.

Case 3. $y=0$ and $x^2-y^2=2 \Rightarrow (x,y)=(\pm\sqrt{2},0)$.

Case 4. $x^2-y^2=-2$ and $x^2-y^2=2$. This is impossible.

| S.P. $(a,b)$ | $f_{xx}(a,b)$ | $f_{yy}(a,b)$ | $f_{xy}(a,b)$ | $D$ | Nature of the points |
|---|---|---|---|---|---|
| $(0,0)$ | 2 | $-2$ | 0 | $-4<0$ | Saddle point |
| $(\pm\sqrt{2},0)$ | $-\frac{4}{e}<0$ | $-\frac{4}{e}$ | 0 | $\frac{16}{e^2}>0$ | Local max for both points |
| $(0,\pm\sqrt{2})$ | $\frac{4}{e}>0$ | $\frac{4}{e}$ | 0 | $\frac{16}{e^2}>0$ | Local min for both points |

Note. 1. $D$ can be computed based on values of $f_{xx}(a,b), f_{yy}(a,b), f_{xy}(a,b)$.

2. Other cases are impossible. We leave this as an exercise.

Remark: For $f(x,y)=(x^2-y^2)e^{-\frac{1}{2}(x^2+y^2)}$, note that $0<e^{-\frac{1}{2}(x^2+y^2)}\le 1$ and the fact that $e^{(x^2+y^2)/2}>\left|x^2-y^2\right|$. Therefore, $-1<f(x,y)<1$. It follows that $f(\pm\sqrt{2},0)=\frac{2}{e}$ is the absolute maximum and $f(0,\pm\sqrt{2})=-\frac{2}{e}$ is the absolute minimum. Otherwise, it exists other local minimum or local maximum.

(d) $f(x,y)=(x^2+y^2-1)(xy+4)=x^3y+4x^2+xy^3+4y^2-xy-4$

Solution. $\begin{cases} f_x=3x^2y+y^3+8x-y \\ f_y=3xy^2+x^3+8y-x \end{cases} \quad \begin{cases} f_{xx}=f_{yy}=6xy+8 \\ f_{xy}=3x^2+3y^2-1 \end{cases}$

$$\begin{cases} f_x=0 \\ f_y=0 \end{cases} \Rightarrow \begin{cases} 3x^2y+y^3+8x-y=0 \ (1) \\ 3xy^2+x^3+8y-x=0 \ (2) \end{cases}$$

$(1)+(2): x^3+y^3+3xy(x+y)+7(x+y)=0$

$\Rightarrow (x+y)(x^2-xy+y^2)+3xy(x+y)+7(x+y)=0 \because x^3+y^3=(x+y)(x^2-xy+y^2)$

$\Rightarrow (x+y)(x^2+2xy+y^2+7)=0$

$\Rightarrow x+y=0$ or $x^2+2xy+y^2+7=0$

Since $x^2+2xy+y^2+7=(x+y)^2+7\geq 7\neq 0$, it follows that $x=-y$.

From (1), we get $-3x^3-x^3+8x+x=0\Rightarrow x(9-4x^2)=0\Rightarrow x=0,\pm\frac{3}{2}$.

$$D=(6xy+8)^2-(3x^2+3y^2-1)^2$$

| S.P. $(a,b)$ | $f_{xx}(a,b)$ | $f_{xy}(a,b)$ | $D$ | Nature of the points |
|---|---|---|---|---|
| (0,0) | $8>0$ | $-1$ | $63>0$ | Local minimum |
| $(-\frac{3}{2},\frac{3}{2}),(\frac{3}{2},-\frac{3}{2})$ | | $\frac{25}{2}$ | $-126<0$ | Saddle points |

Note that $f(5,-1)=-25<f(0,0)=-4$, so $(0,0)$ is not the absolute minimum point.

(e) $f(x,y)=(y-2x)^5+30x^3-15y^3$

Solution. $\begin{cases} f_x=-10(y-2x)^4+90x^2 \\ f_y=5(y-2x)^4-45y^2 \end{cases}$ $\begin{cases} f_{xx}=80(y-2x)^3+180x \\ f_{yy}=20(y-2x)^3-90y \\ f_{xy}=-40(y-2x)^3 \end{cases}$

Write $u=y-2x$. Then

$$\begin{cases} f_x=0 \\ f_y=0 \end{cases}\Rightarrow \begin{cases} -u^4+9x^2=0 \ (1) \\ u^4-9y^2=0 \quad (2) \end{cases} \overset{(1)+(2)}{\Rightarrow} x^2=y^2\Rightarrow x=\pm y.$$

Case 1. $x=y$. Then from (1): $-(x-2x)^4+9x^2=0\Rightarrow 9x^2-x^4=0$

$\Rightarrow (x,y)=(0,0),(3,3),(-3,-3)$

Case 2. $x=-y$. Then from (1): $-(-x-2x)^4+9x^2=0\Rightarrow 9x^2-3^4x^4=0$

$\Rightarrow (x,y)=(0,0),(\frac{1}{3},-\frac{1}{3}),(-\frac{1}{3},\frac{1}{3})$

Now, $D=(80u^3+180x)(20u^3-90y)-(40u^3)^2$.

| $x$ | $y$ | $u$ | $D$ | Nature of the points |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | No conclusion |
| $\frac{1}{3}$ | $-\frac{1}{3}$ | –1 | –1800 | Saddle point |
| $-\frac{1}{3}$ | $\frac{1}{3}$ | –1 | –1800 | Saddle point |
| 3 | 3 | –3 | 145800 | Local max because $f_{xx}(3,3)<0$ |
| –3 | –3 | 3 | 145800 | Local min because $f_{xx}(-3,-3)>0$ |

2. Find the minimum distance of the origin to the plane $x+2y+3z=5$.

Solution. Minimise the objective function $f(x,y,z)=x^2+y^2+z^2$ ; subject to $x+2y+3z=5$. Let $g(x,y,z)=x+2y+3z-5$ so that $g=0$. We apply the method of Lagrange multiplier, $\nabla(f+\lambda g)=0$.

$$\begin{cases} 2x+\lambda=0 \\ 2y+2\lambda=0 \\ 2z+3\lambda=0 \end{cases} \Rightarrow \begin{cases} x=-\frac{1}{2}\lambda \\ y=-\lambda \\ z=-\frac{3}{2}\lambda \end{cases}$$

Sub $x$, $y$ and $z$ into $x+2y+3z=5$ to obtain $(-\frac{1}{2}-2-\frac{9}{2})\lambda=5\Rightarrow\lambda=-\frac{5}{7}$. Thus, $(x,y,z)=(\frac{5}{14},\frac{5}{7},\frac{15}{14})$. To verify that this is the minimum point, we choose $(5,0,0)$ and compare $f(5,0,0)=25>\frac{25}{14}=f(\frac{5}{14},\frac{5}{7},\frac{15}{14})$. Hence, the minimum distance is given by

$$d_{\min}=\sqrt{f(\tfrac{5}{14},\tfrac{5}{7},\tfrac{15}{14})}=\tfrac{5}{\sqrt{14}}.$$

3. Let $f(x,y,z)=4x^2+y^2+5z^2$. Use Lagrange multipliers to find the point on the plane $2x+3y+4z=12$ at which $f$ has its least value.

Solution. Minimise the objective function $f(x,y,z)=4x^2+y^2+5z^2$ subject to $g(x,y,z)=2x+3y+4z-12=0$. Find $x,y,z,\lambda$ such that

$$\nabla\Big(f+\lambda g\Big)=0\Rightarrow\begin{pmatrix}8x+2\lambda\\2y+3\lambda\\10z+4\lambda\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}.$$

We sub $x$, $y$ and $z$ (in term of $\lambda$) into $2x+3y+4z-12=0$ to obtain:

$$-\tfrac{1}{2}\lambda-\tfrac{9}{2}\lambda-\tfrac{8}{5}\lambda=12\Rightarrow\lambda=-\tfrac{20}{11}\Rightarrow(x,y,z)=\Big(\tfrac{5}{11},\tfrac{30}{11},\tfrac{8}{11}\Big).$$

To verify that the point is a minimum point, we choose $(6,0,0)$ compare:

$$f(6,0,0)=144>\tfrac{120}{11}=f\Big(\tfrac{5}{11},\tfrac{30}{11},\tfrac{8}{11}\Big)$$

We conclude that $f_{\min}\Big(\frac{5}{11},\frac{30}{11},\frac{8}{11}\Big)=\frac{120}{11}$.

4. Use Lagrange multipliers to find the point on the sphere $x^2+y^2+z^2=1$ which is closest to the point $(2,3,4)$.

Solution. The objective function is to minimise $f(x,y,z)=(x-2)^2+(y-3)^2+(z-4)^2$ subject to $g(x,y,z)=x^2+y^2+z^2-1=0$. Find $x,y,z,\lambda$ such that

$$\nabla(f+\lambda g)=0 \Rightarrow \begin{cases} 2(x-2)+2x\lambda=0 \\ 2(y-3)+2y\lambda=0 \\ 2(z-4)+2z\lambda=0 \end{cases} \Rightarrow \begin{cases} x=\frac{2}{1+\lambda} \\ y=\frac{3}{1+\lambda} \\ z=\frac{4}{1+\lambda} \end{cases}.$$

Sub $x$, $y$ and $z$ (in term of $\lambda$) in $g=0$ to obtain $\frac{4+9+16}{(1+\lambda)^2}=1 \Rightarrow \lambda=-1\pm\sqrt{29}$.

So, $(x,y,z)=\left(\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}\right),\left(-\frac{2}{\sqrt{29}},-\frac{3}{\sqrt{29}},-\frac{4}{\sqrt{29}}\right)$. Since

$$f\left(\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}\right)<f\left(-\frac{2}{\sqrt{29}},-\frac{3}{\sqrt{29}},-\frac{4}{\sqrt{29}}\right),$$

so we conclude that $\left(\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}\right)$ is the point on the sphere $x^2+y^2+z^2=1$ which is the closest to the point (2, 3, 4).

5. Find the maximum distance from the origin to the curve $5x^2+6xy+5y^2-8=0$.

Solution. Maximise $z=f(x,y)=x^2+y^2$ subject to $0=g(x,y)=5x^2+6xy+5y^2-8$ $=5(x+y)^2-4(xy+2)$. Find $x,y,\lambda$ such that $\nabla(f+\lambda g)=0$

$$\begin{cases} 2x+\lambda(10x+6y)=0 \\ 2y+\lambda(10y+6x)=0 \end{cases} \Rightarrow \begin{cases} (1+5\lambda)x+3\lambda y=0\ (1) \\ (1+5\lambda)y+3\lambda x=0\ (2) \end{cases} \underset{(1)+(2)}{\Rightarrow} (1+8\lambda)(x+y)=0.$$

Case 1. $\lambda=-\frac{1}{8}$. Thus from (1), we have $x=y$. Substitute this into:

$$g(x,y)=5(x+y)^2-4(xy+2)=0, \text{ simplify to get}$$

$$x^2=\tfrac{1}{2} \Rightarrow (x,y)=\left(\tfrac{1}{\sqrt{2}},\tfrac{1}{\sqrt{2}}\right),\left(-\tfrac{1}{\sqrt{2}},-\tfrac{1}{\sqrt{2}}\right).$$

Note that $f\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)=f\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)=1$.

Case 2. $x=-y$. Substitute this into $g(x,y)=0$, we have

$$x^2=2 \Rightarrow (x,y)=(\sqrt{2},-\sqrt{2}),(-\sqrt{2},\sqrt{2}).$$

Note that $f(\sqrt{2},-\sqrt{2})=f(-\sqrt{2},\sqrt{2})=4$.

Hence, the maximum distance, $d_{\max}=\sqrt{f_{\max}}=2$ at $(\sqrt{2},-\sqrt{2})$ or $(-\sqrt{2},\sqrt{2})$.

6. What are the dimensions of the box of maximum volume with sides parallel to the co-ordinate planes which can be contained inside the ellipsoid $x^2+9y^2+4z^2=36$?

Solution. Let $(x,\ y,\ z)$ be the corner of the box which lies in the first octant. The box has sides of length $2x$, $2y$, $2z$, and volume $8xyz$. We maximise volume $V=8xyz$ subject to $x^2+9y^2+4z^2=36$. Find $x,y,z>0$ and $\lambda\neq 0$ such that

$$\begin{pmatrix} 8yz+2x\lambda \\ 8xz+18y\lambda \\ 8xy+8z\lambda \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}\Rightarrow\begin{cases} 4yz+x\lambda=0 & (\times x) \\ 4xz+9y\lambda=0 & (\times y) \\ 4xy+4z\lambda=0 & (\times z) \end{cases}\Rightarrow\begin{cases} 4xyz=-x^2\lambda & (1) \\ 4xyz=-9y^2\lambda & (2) \\ 4xyz=-4z^2\lambda & (3) \end{cases}$$

$$\Rightarrow\begin{cases} (1)=(2):(x^2-9y^2)\lambda=0 \\ (1)=(3):(x^2-4z^2)\lambda=0 \\ (2)=(3):(9y^2-4z^2)\lambda=0 \end{cases}$$

Note that $\lambda\neq 0$ (or otherwise $x=y=z=0$). Thus, we have $x^2=9y^2=4z^2$, which mean $x^2+9y^2+4z^2=36\Rightarrow 3x^2=36\Rightarrow(x,y,z)=(2\sqrt{3},\frac{2}{3}\sqrt{3},\sqrt{3})$. So, the dimensions (width, length and height) of the box are $4\sqrt{3}$, $\frac{4}{3}\sqrt{3}$, $2\sqrt{3}$ so that the max volume is $V_{\max}=32\sqrt{3}$. Verify that $V(3\sqrt{3},\frac{1}{3}\sqrt{3},\sqrt{\frac{3}{2}})=8(3)\sqrt{\frac{3}{2}}<32\sqrt{3}$.

7. Find the maximal and minimal values of $f(x,y)=\frac{x}{a}+\frac{y}{b}, a,b\in\mathbb{R}^+$ with constraint $x^2+y^2=1$.

Solution. Let $g(x,y)=x^2+y^2-1$ so that $g=0$. Find $x,y$ and $\lambda\neq 0$ such that

$$\nabla\big(f+\lambda g\big)=0\Rightarrow\begin{pmatrix} \frac{1}{a}+\lambda(2x) \\ \frac{1}{b}+\lambda(2y) \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}\Rightarrow\begin{cases} x=-\frac{1}{2a\lambda} \\ y=-\frac{1}{2b\lambda} \end{cases}.$$

Sub $x$ and $y$ in $g(x,y)=x^2+y^2-1=0$: $\frac{1}{4a^2\lambda^2}+\frac{1}{4b^2\lambda^2}=1\Rightarrow\lambda^2=\frac{a^2+b^2}{4a^2b^2}\Rightarrow\lambda=\pm\frac{\sqrt{a^2+b^2}}{2ab}$ (no need $\sqrt{a^2b^2}=|ab|$ since $a,b>0$). Hence, $(x,y)=\left(\frac{b}{\sqrt{a^2+b^2}},\frac{a}{\sqrt{a^2+b^2}}\right),\ \left(-\frac{b}{\sqrt{a^2+b^2}},-\frac{a}{\sqrt{a^2+b^2}}\right)$.

$$\begin{cases} f\left(\frac{b}{\sqrt{a^2+b^2}},\frac{a}{\sqrt{a^2+b^2}}\right)=\frac{b}{a\sqrt{a^2+b^2}}+\frac{a}{b\sqrt{a^2+b^2}}=\frac{b^2+a^2}{ab\sqrt{a^2+b^2}}=\frac{\sqrt{a^2+b^2}}{ab}=f_{\max} \\ f\left(-\frac{b}{\sqrt{a^2+b^2}},-\frac{a}{\sqrt{a^2+b^2}}\right)=-\left(\frac{b}{a\sqrt{a^2+b^2}}+\frac{a}{b\sqrt{a^2+b^2}}\right)=-\frac{\sqrt{a^2+b^2}}{ab}=f_{\min} \end{cases}.$$

8. Determine the maximum and minimum of $f(x,y) = 4x^2 - xy + y^2 + y$ in the region $x^2 + \frac{1}{4}y^2 \le 1$.

Solution. We find the max/min of $f$ with no restriction imposed on $f$, hoping that the answer fulfils $x^2 + \frac{1}{4}y^2 \le 1$. We then apply the method of *Lagrange multiplier* with constraint $g(x,y) = x^2 + \frac{1}{4}y^2 - 1 = 0$ to look for other possible answer.

$$f(x,y) = (2x - \tfrac{1}{4}y)^2 + \tfrac{15}{16}(y^2 + \tfrac{8}{15})^2 - \tfrac{4}{15} \ge -\tfrac{4}{15} \text{ when } (x,y) = \left(-\tfrac{1}{15}, -\tfrac{8}{15}\right).$$

Check that the point satisfies $x^2 + \frac{1}{4}y^2 \le 1$: $\frac{1}{225} + \frac{1}{4}\left(\frac{64}{225}\right) = \frac{17}{225} < 1$.

*Lagrange multiplier*. We search only the maximum. Let $g(x,y) = x^2 + \frac{1}{4}y^2 - 1$.

$$\nabla\left(f + \lambda g\right) = 0 \Rightarrow \begin{cases} 8x - y + 2\lambda x = 0 \\ -x + 2y + 1 + \frac{1}{2}\lambda y = 0 \end{cases} \Rightarrow \begin{cases} 2x(4+\lambda) = y & (1) \\ y(\lambda + 4) + 2 = 2x & (2) \end{cases}$$

Sub (1) into (2): $\frac{y^2}{2x} + 2 = 2x \Rightarrow y^2 = 4x(x-1)$. Sub this into $g(x,y) = x^2 + \frac{1}{4}y^2 - 1 = 0$:

$$x^2 + x(x-1) - 1 = 0 \Rightarrow 2x^2 - x - 1 = 0 \Rightarrow x = 1, -\tfrac{1}{2} \Rightarrow y = 0, \pm\sqrt{3}.$$

Evaluate $\quad f(-\frac{1}{2}, -\sqrt{3}) = \frac{8-3\sqrt{3}}{2} < f(1,0) = 4 < f(-\frac{1}{2}, \sqrt{3}) = \frac{8+3\sqrt{3}}{2}$

We conclude that $f_{\max} = f(-\frac{1}{2}, \sqrt{3}) = \frac{8+3\sqrt{3}}{2}$; and $f_{\min} = f(-\frac{1}{15}, -\frac{8}{15}) = -\frac{4}{15}$.

**Extra Exercises**

9. (a) Maximise $\sum_{i=1}^{n} x_i y_i$ subject to $\sum_{i=1}^{n} x_i^2 = 1$ and $\sum_{i=1}^{n} y_i^2 = 1$.

(b) Hence, prove $\sum_{i=1}^{n} a_i b_i \le \sqrt{\sum_{i=1}^{n} a_i^2 \sum_{i=1}^{n} b_i^2}$ (this is called the Cauchy's inequality).

Proof. (a) Let $f(x_1,...,x_n,y_1,...,y_n) = \sum_{i=1}^{n} x_i y_i$, $g(x_1,...,x_n,y_1,...,y_n) = \sum_{i=1}^{n} x_i^2 - 1$ and

$$h(x_1,...,x_n,y_1,...,y_n) = \sum_{i=1}^{n} y_i^2 - 1 \text{ so that } g = 0 = h.$$

Now, $\nabla f = (y_1,...,y_n,x_1,...,x_n), \nabla g = (2x_1,...,2x_n,0,...,0), \nabla h = (0,...,0,2y_1,...,2y_n)$

$$\nabla(f + \lambda g + \mu h) = 0 \Rightarrow \begin{cases} y_i + 2\lambda x_i = 0 \\ x_i + 2\mu y_i = 0 \end{cases} \forall i = 1,...,n.$$

Clearly $\lambda, \mu \neq 0$. Otherwise, $x_i = y_i = 0$ $\forall i = 1,...,n$ contradicts to $g = h = 0$.

Now $1 = \sum_{i=1}^{n} y_i^2 = \sum_{i=1}^{n} 4\lambda^2 x_i^2 = 4\lambda^2 \sum_{i=1}^{n} x_i^2 = 4\lambda^2 \Rightarrow \lambda = \pm\frac{1}{2}$, and similarly $\mu = \pm\frac{1}{2}$.

Thus, $y_i \pm x_i = 0$ and $x_i \pm y_i = 0$. Since $x_i, y_i \neq 0$ $\forall i = 1,...,n$, it follows that $x_i = y_i$ $\forall i = 1,...,n$. So $f(x_1,...,x_n,y_1,...,y_n) = \sum_{i=1}^{n} x_i^2 = 1$.

(b) Let $A = \sqrt{\sum_{i=1}^{n} a_i^2}$ and $B = \sqrt{\sum_{i=1}^{n} b_i^2}$ such that $A, B > 0$ (if $A = B = 0$, then the result is obviously true). Let $x_i = \frac{a_i}{A}, y_i = \frac{b_i}{B}$. Then $\sum_{i=1}^{n} x_i^2 = 1 = \sum_{i=1}^{n} y_i^2$.

By (a), $\sum_{i=1}^{n} x_i y_i \le 1$. Hence, $\sum_{i=1}^{n} \left(\frac{a_i}{A} \cdot \frac{b_i}{B}\right) \le 1$ or $\sum_{i=1}^{n} a_i b_i \le AB$.

10. Find the maximum and minimum of $f(x,y) = xy$ if $x^2 + y^2 \le 1$.

Solution. $\begin{cases} (x-y)^2 \ge 0 \Rightarrow 1 \ge x^2 + y^2 \ge 2xy \Rightarrow xy \le \frac{1}{2} \\ (x+y)^2 \ge 0 \Rightarrow 1 \ge x^2 + y^2 \ge -2xy \Rightarrow xy \ge -\frac{1}{2} \end{cases}$.

So, $-\frac{1}{2} \le f(x,y) \le \frac{1}{2}$, and $f_{\min} = -\frac{1}{2}$ when $(x,y) = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}), (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$;

and $f_{\max} = \frac{1}{2}$ when $(x,y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$.

11. Optimise $f(x,y)=x^2+2y^2-xy+x+1$ subject to $x^2+y^2\le 1$ using the KKT method.

Solution. (Minimise) Let $g(x,y)=x^2+y^2-1\le 0$. Then

$$\nabla\big(f+\mu g\big)=0\Rightarrow\begin{pmatrix}2x-y+1+2\mu x\\ 4y-x+2\mu y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow\begin{cases}2x-y+1=-2\mu x & (1)\\ 4y-x=-2\mu y & (2)\\ \mu\ge 0 & (3)\\ \mu(x^2+y^2-1)=0 & (4)\end{cases}$$

From (4), it implies that either $\mu=0$ or $x^2+y^2=1$.

Case 1. $\mu=0$. Then $2x-y=-1$ and $4y=x$ from (1) and (2). So, $(x,y)=(-\frac{4}{7},-\frac{1}{7})$.

Case 2. $\mu=0$ and $x^2+y^2=1$. Then $(x,y)=(-\frac{4}{7},-\frac{1}{7})$ from $\mu=0$, a contradiction because $\left(-\frac{4}{7}\right)^2+\left(-\frac{1}{7}\right)^2\neq 1$.

Case 3. $x^2+y^2=1$ (5). Take $(1)\times y=(2)\times x$:

$$2xy-y^2+y=4xy-x^2\Rightarrow 1=x^2+y^2=2y^2+2xy-y$$

$$\text{or } 2y^2+(2x-1)y-1=0\Rightarrow y=\frac{(1-2x)\pm\sqrt{(2x-1)^2+8}}{4}\ (6).$$

Sub (6) to (5) and solve:

When $x^2+\left(\frac{(1-2x)+\sqrt{(2x-1)^2+8}}{4}\right)^2=1\Rightarrow x=0,0.8269434246$.

When $x^2+\left(\frac{(1-2x)-\sqrt{(2x-1)^2+8}}{4}\right)^2=1\Rightarrow x=0.6311094891,-0.958052913$.

When $x=0$, $y=\pm 1$; and when $x=0.827$, $y=\pm 0.562$.

When $x=0.6311$, $y=\pm 0.77569$; and when $x=-0.958053$, $y=\pm 0.28659$.

| $x$ | $y$ | $\mu=\frac{y-2x-1}{2x}$ | $\mu=\frac{x-4y}{2y}$ | Contradiction? |
|---|---|---|---|---|
| 0 | 1 | Not applicable | – 2 | Yes |
| 0 | – 1 | Not applicable | – 2 | Yes |
| 0.827 | 0.562 | $<0$ | $<0$ | Yes |
| 0.827 | – 0.562 | $<0$ | $<0$ | Yes |
| 0.6311 | 0.77569 | $<0$ | $<0$ | Yes |
| 0.6311 | – 0.77569 | $<0$ | $<0$ | Yes |
| – 0.95805 | 0.28659 | $<0$ | $<0$ | Yes |
| – 0.95805 | – 0.28659 | $<0$ | $<0$ | Yes |
| $-\frac{4}{7}$ | $-\frac{1}{7}$ | 0 | 0 | No |

All results obtained in Case 3 are not possible. Conclusion: $f\left(-\frac{4}{7},-\frac{1}{7}\right)=\frac{5}{7}=f_{\min}$ because if we take any point fulfilling $x^2+y^2\le 1$, such as $(x,y)=(0,0)$, we have $f(0,0)=1>\frac{5}{7}=f_{\min}$.

(Maximise) Let $g(x,y) = x^2 + y^2 - 1 \leq 0$ .Then

$$\nabla\big(f - \mu g\big) = 0 \Rightarrow \begin{pmatrix} 2x - y + 1 - 2\mu x \\ 4y - x - 2\mu y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2x - y + 1 = 2\mu x & (1) \\ 4y - x = 2\mu y & (2) \\ \mu \geq 0 & (3) \\ \mu(x^2 + y^2 - 1) = 0 & (4) \end{cases}$$

We obtained all the point and verify the value of $\mu$ from (1) and (2).

| $x$ | $y$ | $\mu = \frac{1-y+2x}{2x}$ | $\mu = \frac{4y-x}{2y}$ | Contradiction? | $f\,(x,y)$ | Remark |
|---|---|---|---|---|---|---|
| 0 | 1 | - | 2 | No | 3 | |
| 0 | – 1 | - | 2 | Yes | | |
| 0.826943 | 0.562285 | 1.2646583 | 1.2646583 | No | 2.68 | |
| 0.826943 | – 0.56229 | 1.9446143 | 2.7353417 | Yes | | |
| 0.631109 | 0.775694 | 1.1777079 | 1.5931967 | Yes | | |
| 0.631109 | – 0.77569 | 2.4068033 | 2.4068033 | No | 3.722 | Max |
| – 0.95805 | 0.286591 | 0.6276779 | 3.6714615 | Yes | | |
| – 0.95805 | – 0.28659 | 0.3285385 | 0.3285385 | No | 0.85 | |
| $-\frac{4}{7}$ | $-\frac{1}{7}$ | 0 | 0 | No | $\frac{5}{7}$ | |

Hence, $f(0.631,-0.776) = 3.722$ is the maximum.

**Past year test/final exam questions**

12. Let $f(x,y) = x^3 + x^2 + 2xy - 3x + y^2$.

(a) Find all the stationary points of $f(x,y)$.

(b) Determine the nature of the stationary points.

(c) Is the local extremum obtained in (b) an absolute extremum? Explain.

Solution. (a) $f_x = 3x^2 + 2x + 2y - 3, f_y = 2x + 2y, f_{xx} = 6x + 2, f_{yy} = 2 = f_{xy} = f_{yx}$.

$$3x^2 + 2x + 2y - 3 = f_x = 0 = f_y = 2x + 2y \Rightarrow x = -y$$
$$\Rightarrow 3x^2 + 2x - 2x - 3 = 0 \Rightarrow x = \pm 1 \Rightarrow (x,y) = (1,-1),(-1,1)$$

(b) $D(x,y) = 2(6x+2) - 4$.

Now $D(1,-1) = 2(6+2) - 4 > 0$ and $f_{xx}(1,-1) = 8 > 0$. So, $(1,-1)$ is a local min.

While $D(-1,1) = 2(-6+2) - 4 < 0$. So, $(-1,1)$ is a saddle point.

Alternative

$$0 = \det\begin{pmatrix} 6x+2-\lambda & 2 \\ 2 & 2-\lambda \end{pmatrix} \Rightarrow 0 = (6x+2-\lambda)(2-\lambda) - 4 \Rightarrow 0 = (2-\lambda)^2 + 6x(2-\lambda) - 4$$

$0 = (2-\lambda)^2 + 6(2-\lambda) - 4 \Rightarrow \lambda = 5 \pm \sqrt{13} > 0$. So $(1,-1)$ is a local min.

$0 = (2-\lambda)^2 - 6(2-\lambda) - 4 \Rightarrow u^2 - 6u - 4 = 0 \Rightarrow \lambda = -1 \pm \sqrt{13}$ where one is positive and one is negative. So $(-1,1)$ is a saddle point.

(c) $f(1,-1) = -2 > f(-5,-5) = -10$ so $(1,-1)$ is not an absolute min point.

13. Find the minimum distance between $x + 2y + z = 4$ and the origin $(0,0,0)$.

Solution. Minimise $f(x,y,z) = x^2 + y^2 + z^2$ subject to constraint $g(x,y,z) = x + 2y + z - 4 = 0$. Using Lagrange multiplier,

$$\nabla f = \nabla \lambda g \Rightarrow \begin{cases} 2x = \lambda \\ 2y = 2\lambda \Rightarrow \lambda = 2x = y = 2z. \\ 2z = \lambda \end{cases}$$

Sub into $g(x,y,z) = x + 2y + z - 4 = 0$ to obtain

$$\tfrac{1}{2}\lambda + 2\lambda + \tfrac{1}{2}\lambda = 4 \Rightarrow \lambda = \tfrac{4}{3} \Rightarrow (x,y,z) = \left(\tfrac{2}{3},\tfrac{4}{3},\tfrac{2}{3}\right).$$

Take $(0,2,0)$ (since $g(0,2,0) = 0$) and note that $f(0,2,0) = 4 > f\left(\tfrac{2}{3},\tfrac{4}{3},\tfrac{2}{3}\right) = \tfrac{24}{9}$.

Therefore, $f_{\min} = \tfrac{24}{9}$. Hence, the minimum distance is $d_{\min} = \sqrt{\tfrac{8}{3}}$ at $\left(\tfrac{2}{3},\tfrac{4}{3},\tfrac{2}{3}\right)$.

Alternative. Sub $z = 4 - x - 2y$ into $f(x,y,z) = x^2 + y^2 + z^2$ to obtain $f(x,y) = x^2 + y^2 + (4 - x - 2y)^2$ and use $D$-test.

$f_x(x,y) = 2x - 2(4 - x - 2y) = 0$ and $f_y(x,y) = 2y - 4(4 - x - 2y) = 0$.

So, $x = 2 - y$ and $5y = 8 - 2x$ to get $x = \frac{2}{3}, y = \frac{4}{3}$ and $z = \frac{2}{3}$.

Now $f_{xx}(x,y) = 4 > 0, f_{yy}(x,y) = 10, f_{xy}(x,y) = 4 \Rightarrow D(x,y) = 40 - 16 > 0$ for any $(x,y)$, so $\left(\frac{2}{3}, \frac{4}{3}, \frac{2}{3}\right)$ is a local min. Hence, $f_{\min} = \frac{24}{9} = \frac{8}{3} \Rightarrow d_{\min} = \sqrt{\frac{8}{3}}$ but this does not prove that $\left(\frac{2}{3}, \frac{4}{3}, \frac{2}{3}\right)$ is the absolute min, to do so, we complete the square:

$$\begin{aligned} f(x,y) &= x^2 + y^2 + (4 - x - 2y)^2 \\ &= x^2 + y^2 + x^2 + 4y^2 + 16 - 8x - 16y + 4xy \\ &= 2x^2 + 4xy - 8x - 16y + 5y^2 + 16 \\ &= 2(x-2)^2 + 4(x-2)y + 2y^2 - 8y + 3y^2 + 8 \\ &= 2(x + y - 2)^2 + 3y^2 - 8y + 8 \\ &= 2(x + y - 2)^2 + 3\left(y^2 - \tfrac{8}{3}y + \tfrac{16}{9}\right) + 8 - \tfrac{16}{3} \\ &= 2(x + y - 2)^2 + 3\left(y - \tfrac{4}{3}\right)^2 + \tfrac{8}{3} \geq \tfrac{8}{3} \qquad \text{when } (x,y,z) = \left(\tfrac{2}{3}, \tfrac{4}{3}, \tfrac{2}{3}\right). \end{aligned}$$

14. By using the method of Lagrange multipliers, find the maximum and minimum of the function $f(x,y,z) = x + 3y + 2z$ subject to the condition $x^2 + y^2 + z^2 = 14$.

Solution. Let $g(x,y,z) = x^2 + y^2 + z^2 - 14$. Find $x, y, z, \lambda$ from

$$\nabla(f + \lambda g) = 0 \text{ and } g(x,y,z) = 0.$$

$$\begin{cases} 1 + 2\lambda x = 0 \\ 3 + 2\lambda y = 0 \\ 2 + 2\lambda z = 0 \end{cases} \Rightarrow \begin{cases} x = -\frac{1}{2\lambda} \\ y = -\frac{3}{2\lambda} \\ z = -\frac{1}{\lambda} \end{cases} \Rightarrow \frac{1}{\lambda^2}\left(\tfrac{1}{4} + \tfrac{9}{4} + 1\right) = 14 \Rightarrow \lambda = \pm\tfrac{1}{2}$$

($\lambda \neq 0$ or else $\nabla(f + \lambda g) \neq 0$, a contradiction)

So, $(x,y,z) = (1,3,2), (-1,-3,-2)$

Now, $f(1,3,2) = 14 = f_{\max}$ and $f(-1,-3,-2) = -14 = f_{\min}$.